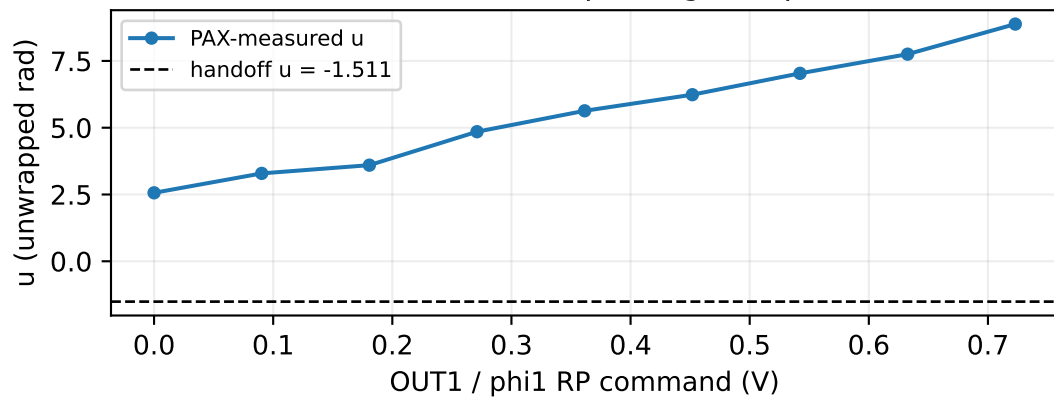
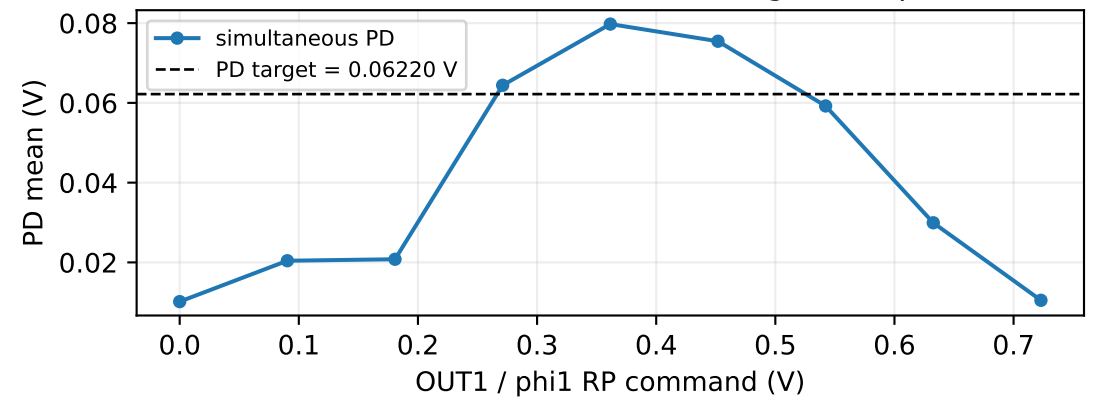


Phi1 PD FPGA lock — 005148_phi1-pd-lock-test_phi1-pd-lock-3

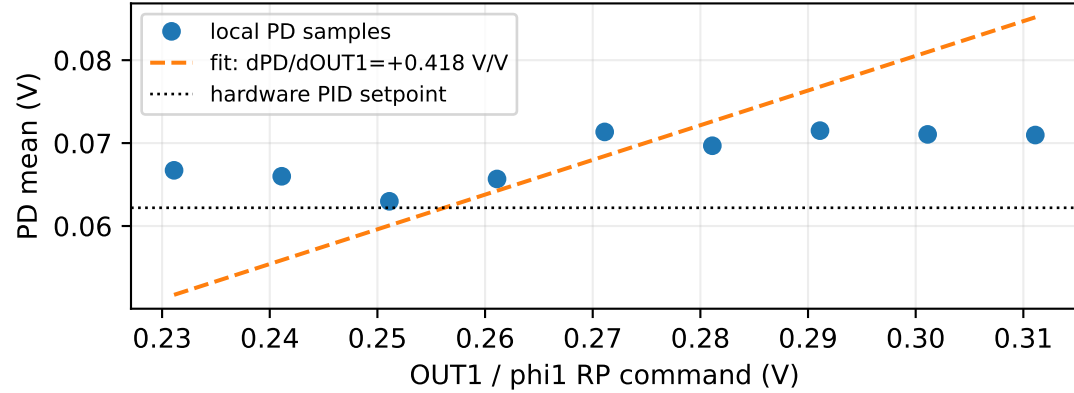
PAX-assisted ϕ_1 rough map



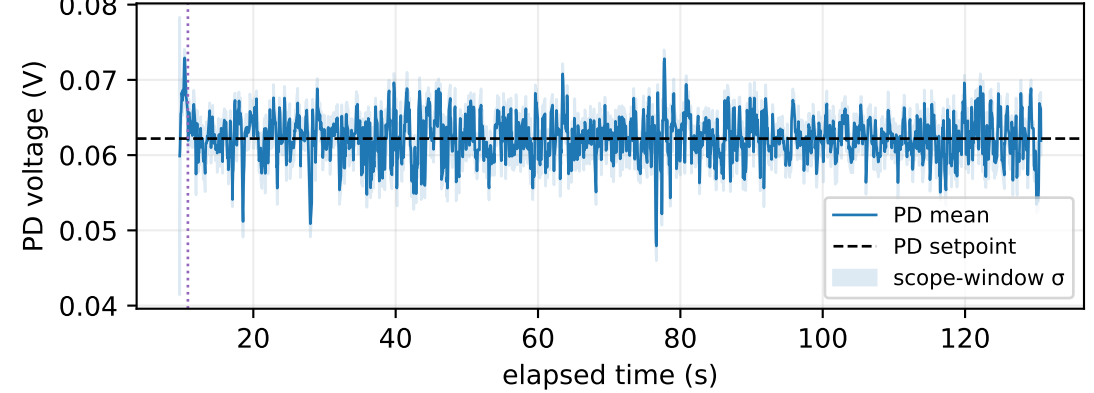
PD value associated with rough u map



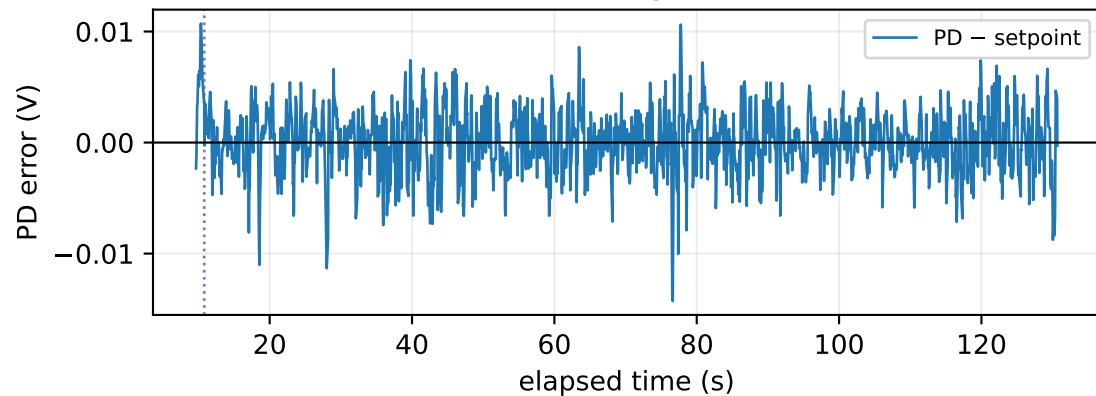
Automatic local PD calibration



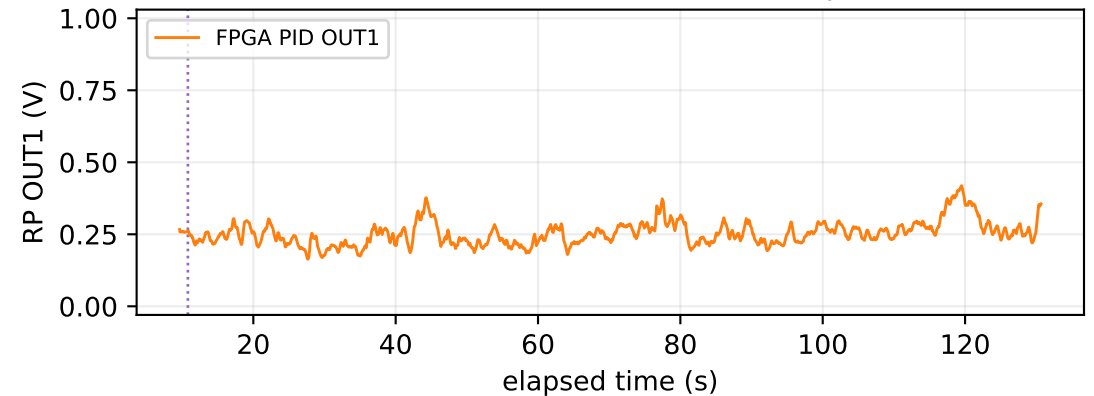
FPGA PD lock



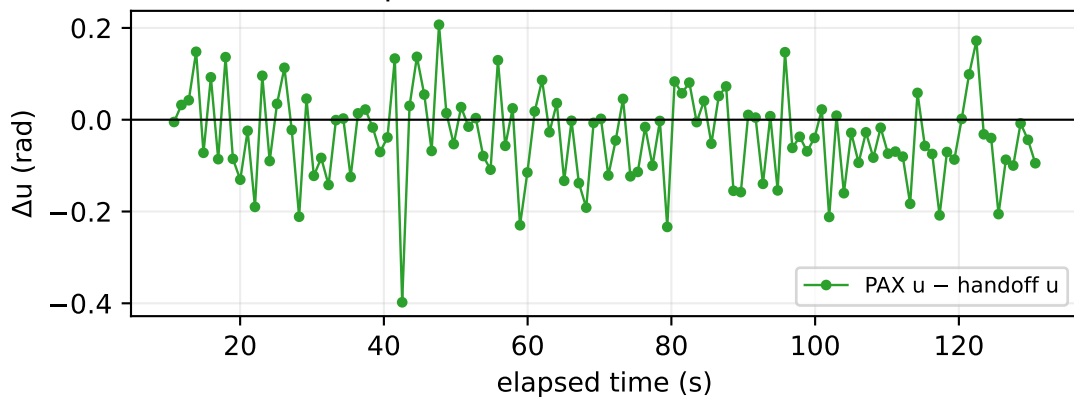
PD locking error



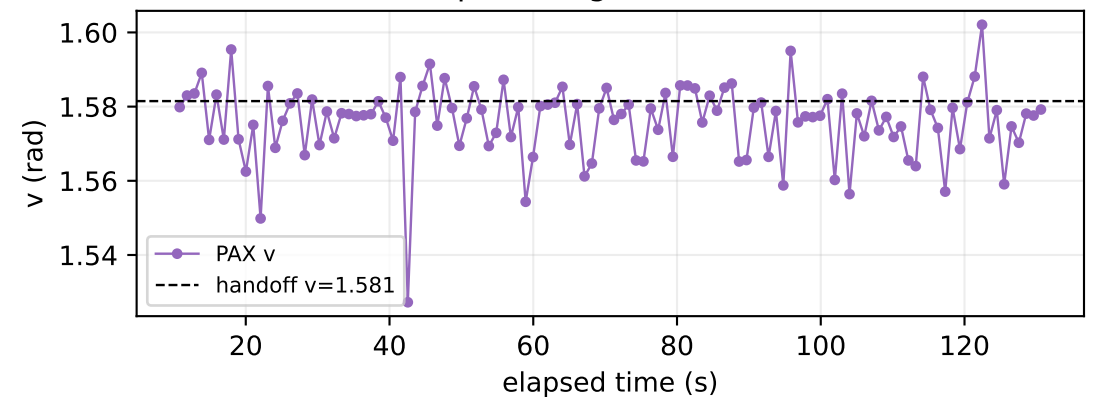
Continuous FPGA actuator output



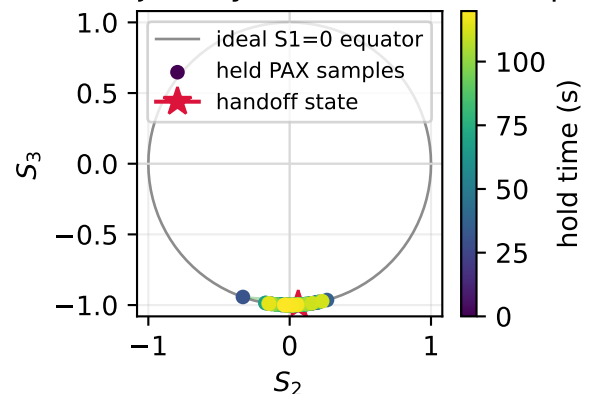
Independent PAX azimuth validation



PAX polar-angle cross-check



Held lock trajectory on the Poincare equator



Interpretation

- Tight cluster near the red star: PD lock preserves the selected u branch.
- Motion around the circle: residual ϕ_1/u drift or branch movement.
- Radial motion: polarization is leaving the ideal equator.

PAX handoff target $u = -1.5105$ rad PD setpoint = 0.06220 V local slope = +0.4184 V/V
FPGA PID: P = -0.358, I = -5 Hz Duration = 121.0 s median |PD error| = 0.00180 V (late 0.00180 V) OUT1 range = 0.164...0.419 V PAX median $|\Delta u| = 0.070$ rad (118 snapshots)